Practical 2a: Function Overloading

AIM : To demonstrate function overloading in C# by creating multiple methods with the same name but different parameters.

PROGRAM CODE

```
using System;

namespace FunctionOverloading
{
   class Calculator
   {
      // Overload 1: Two integers
      public int Add(int a, int b)
      {
         Console.WriteLine($"Add(int, int) called");
         return a + b;
      }

      // Overload 2: Three integers
      public int Add(int a, int b, int c)
      {
         Console.WriteLine($"Add(int, int, int) called");
         return a + b + c;
      }

      // Overload 3: Two doubles
      public double Add(double a, double b)
      {
         Console.WriteLine($"Add(double, double) called");
         return a + b;
      }

      // Overload 4: String concatenation
      public string Add(string a, string b)
      {
         Console.WriteLine($"Add(string, string) called");
         return a + b;
      }
```

```
    }

    class Program
    {
        static void Main(string[] args)
        {
            Calculator calc = new Calculator();
            Console.WriteLine("=== Function Overloading Demo ===");
            Console.WriteLine(calc.Add(10, 20));          // 30
            Console.WriteLine(calc.Add(10, 20, 30));      // 60
            Console.WriteLine(calc.Add(1.5, 2.5));        // 4.0
            Console.WriteLine(calc.Add("Hello, ", "World!"));
            Console.ReadKey();
        }
    }
}
```

Output:

```
=== Function Overloading Demo ===
Add(int, int) called
30
Add(int, int, int) called
60
Add(double, double) called
4
Add(string, string) called
Hello, World!
```